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DATA-DRIVEN FORECASTING OF PRODUCTION RATES IN PETROLEUM RESERVOIRS USING MACHINE LEARNING

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Abstract. *Forecasting production reliably and anticipating changes in rock-fluid system behaviors are pivotal challenges in petroleum reservoir engineering. This work addresses these challenges through a data-driven approach using machine learning methods. The objective is to develop a methodology for forecasting production rates using basic data, such as historical volumes produced and injected, without relying on geological models, fluid properties, well completions, or flow systems information. A novel aspect of our approach is its ability to capture temporal and volumetric relationships between individual wells, and to forecast oil, gas, and water rates using future injection rate schedules as inputs. We employed supervised learning techniques—specifically, linear regression and artificial neural networks—to assess their performance and complexity in meeting our needs. The forecasting accuracy was evaluated using synthetic data from a realistic simulation model. Performance was measured using quantitative metrics, and qualitatively by observing production rate trends at different times in relation to the injection schedules. The results demonstrate a robust capability to replicate trends in fluid production even under new injection strategies, suggesting that this workflow could effectively support reservoir management with the goal of maximizing oil recovery.*

Keywords: *machine learning, petroleum reservoirs, forecasting, decision support*

1. INTRODUCTION

Petroleum reservoirs are complex thermophysical systems where multiphase flow, heat transfer, and pressure-volume-temperature (PVT) relationships govern fluid behavior. Accurately forecasting production rates remains a central challenge in petroleum reservoir engineering. The ability to estimate future production reliably is crucial for both short- and long-term activities, including operational planning, equipment and facility dimensioning, scheduling offloads, evaluating the economic viability of future projects, and estimating reserves. Additionally, anticipating depreciatory reservoir behaviors prone to production loss is essential for implementing mitigating and preventive actions within the scope of reservoir management.

The complexity of production forecasting arises from the inherent difficulty in precisely characterizing *in situ* reservoir properties and accounting for the coupled thermal and flow phenomena, resulting in forecasts that are approximations based on simplified representations of the rock-fluid system. Over decades of development in the industry, several methodologies have been presented and improved to address this challenge. Traditional methods, well-established and widely proven, represent one avenue of approach. In contrast, methodologies leveraging data and machine learning have gained prominence more recently, offering tailored frameworks for specific applications. In this context, we present a contextualization and demonstrate how our novel data-driven methodology contributes to practical applications in reservoir and production management.

1.1 Traditional forecasting methods in reservoir engineering

Traditional forecasting methods in petroleum reservoir engineering encompass various categories, each distinguished by complexity, reliance on prior data, computational demands, and ability to represent phenomena comprehensively.

These methods, extensively explored in literature including textbooks by Ahmed (2005), Dake (1998), and Fanchi (2006), are overviewed below.

- Decline curve analysis (Arps 1945) stands as a seminal analytical method for production forecasting. It involves fitting a decline equation to historical production data, applicable to individual wells or entire reservoirs. The generalized decline model, often represented by the hyperbolic curve (Eq. 1), can also encompass harmonic and exponential variants, as deemed suitable for specific scenarios (Thakur 2017).

$$q_t = \frac{q_i}{(1+b \cdot d_i \cdot t)^{1/b}} \quad (1)$$

To determine the parameters and calculate the production rate at a given time q_t, we must specify the initial rate q_i, the decline rate d_i, and the factor b (where $0<b<1$) to minimize deviations from a reference curve derived from well history or analogous cases. In certain scenarios, as unconventional shale reservoirs, hybrid declines may be appropriate, as reservoirs often tend toward exponential decline (b=0) after a certain period (Belyadi et al. 2019). Due to its simplicity, this method is favored in mature fields where production trends are well-established. However, significant alterations in production conditions, such as pressures below saturation, introduction of new injections, or water/gas breakthroughs, may not be accurately captured.

- The material balance analysis represents a more comprehensive analytical method tracing back to the seminal work of Schilthuis in the early 20th century, who proposed an equation relating produced and injected volumes at a given static pressure (Dake 1998). This equation, based on measured volumes at surface conditions adjusted to reservoir pressure, encompasses rock compressibility and factors such as gas cap presence and aquifer water influx. By expressing the equation as a straight line, as introduced by Havlena and Odeh (1963), production and pressure data can be matched to derive an analytical model of the reservoir. This methodology proves useful for forecasting as long as the drive mechanism remains unchanged; any deviation from the straight line indicates a change in the drive mechanism. This method can be effectively applied to simplified shoebox-type models, where the reservoir is represented as a single block with spatially uniform properties. Effective application requires production history, static pressure data, PVT properties, and petrophysical information such as pore volume and compressibility. However, this simplified representation limits applicability to reservoirs with small-to-medium volumes and minimal heterogeneity.
- Numerical simulation marks a significant advancement in complexity. It involves discretized reservoir representation through grids comprising millions of cells populated with rock and fluid properties estimated from core analysis, well logging, formation tests, and seismic data. Once established, numerical solutions simulate porous flow, mass conservation, and equation-of-state laws at each time step. Accuracy depends critically on data quality; refined grids without representative data may not improve forecasts. Following production periods, models can be refined through history matching (Fanchi 2006), though this poses complex inverse problems where infinite parameter combinations may yield history matches without guaranteeing predictive accuracy. For large, complex reservoirs, modeling and simulation can be costly in data acquisition, computational demands, and specialist time. Simulation runs may extend from hours to days even with high-capacity clusters, what poses a frequent and significant challenge in the oil and gas industry, as documented by Bogachev et al. (2018), and Rios et al. (2020). A common approach to address this challenge is scaling down the model, enlarging grid cells to reduce simulation times, albeit at the expense of reduced result accuracy. These limitations create opportunities for alternative approaches, particularly machine learning-based methods, which can bridge gaps by learning complex patterns directly from data without requiring detailed physical models, offering faster predictions while capturing non-linear relationships that traditional methods may overlook.

1.2 Data-driven methods in reservoir engineering

Data-driven models are increasingly prevalent in the oil and gas industry (Pandey 2020), particularly in scenarios where detailed information on fluid configurations and pressures within the reservoir is not imperative or when a substantial database of production/injection volumes and well sensor data is available, but petrophysical and/or PVT data are lacking. These methods are not intended to replace traditional approaches but rather to complement them, addressing specific gaps.

The use of data-driven tools in reservoir management is of significant importance, particularly in supporting routine decisions such as changing zones, production and injection rate management, and well prioritization during production platform bottlenecks or resource constraints. Extracting knowledge from data can enhance reservoir management in the short and medium term if the developed tools are appropriate.

In the literature, a plethora of data-driven and machine learning methods are applied to forecasting challenges in reservoir engineering, reflecting their increasing prevalence and reliability. These methods exhibit substantial variation depending on factors such as the forecast horizon, selected variables, learning methodology, and ultimate objective.

For instance, Kubota and Reinert (2019) investigated full-field data-driven production forecasting in an onshore mature field subjected to joint water and steam injection. Their study introduced two distinct data-driven approaches – an adaptive linear model and a recurrent neural network (RNN) – utilizing solely total injection rate and the number of operational production wells as inputs to predict total oil rate months in advance.

Davtyan et al. (2020) conducted a study aimed at full-field production forecasting, focusing on evaluating results over a six-month period. Their research explored the influence of four classes of features: integral, encompassing total numbers of producing and injecting wells; local, evaluating distances between producers and injectors; pressures, reflecting Darcy's Law principles; and autoregressive, incorporating temporal windows to account for reservoir "inertia". To ensure user comprehensibility, the authors opted for simple learning methods, such as linear regressions.

Ning et al. (2022) centered their approach on time series methods—ARIMA, LSTM, and the Prophet model—to forecast the oil production rate of a given well as the sole variable on a monthly basis. Their methodology focused on cases where production curves followed typical decline patterns, albeit with inherent noise stemming from operational and measurement issues.

Chakra et al. (2013) employed neural networks (NN) with multiple layers to forecast monthly cumulative oil production for specific wells. Their case studies utilized inputs such as productions of related wells, considering cumulative volumes from one or two previous months, and varying between solely oil volumes or including gas and water production from each well. The model provided estimates of oil production for a single future time step based on these inputs.

In pursuit of fast and accurate production forecasts, Liu et al. proposed a learning paradigm featuring a long short-term memory (LSTM) neural network based on ensemble empirical mode decomposition (EEMD) (Liu et al. 2020). Their methodology focused solely on oil rate time series, selecting appropriate decomposed components for forecasting based on current reservoir conditions and production system status.

We also highlight studies with data-driven forecasts for purposes other than obtaining a time series for flow rates.

Chahar et al. (2022) investigated the estimation of instantaneous flow rates using data from well sensors and flow control parameters, employing gradient boosting and random forests methods. Wang et al. (2019) utilized deep learning techniques to optimize parameters for stimulation operations, aiming to maximize cumulative production after a given time. Jia et al. (2020) explored the optimization of parameters in a waterflood project through machine learning techniques, offering insights into injection efficiency in various reservoir layers. Their findings contributed to defining optimal injection schedules in complex geological settings.

Deng and Pan (2021) and Haghshenas et al. (2021) both employed data-driven modeling to create proxies that can be used replacing a simulation in specific tasks. The former uses them to forecast water breakthrough, and the latter to optimize injection rates aiming to increase recovery factors. Temirchev et al. (2020) proposed a proxy model based on an autoencoder deep neural network trained with multiple simulation model realizations of the simulation model, capable of retaining useful information of the multiphase flow dynamics in the reservoir while requiring only a fraction of the processing time.

Cao et al. (2022) compiled lessons learned from various studies in reservoir engineering, providing insights into the advantages, disadvantages, and limitations of popular data-driven forecasting approaches, which serves as a valuable resource.

In this work, we aim to address a gap in this area of research by proposing a framework based solely on production and injection data, which includes the following key features:

- segmented forecasting by well or for the total of an area or field;
- forecasting of oil, gas, and water rates;
- extraction of information from volumetric and temporal relationships among wells;
- inclusion of schedules of injection rates and operational events.

We believe that these features are essential for the framework to be effectively applied in complex real-world situations. Our goal is to utilize this framework in the analysis of medium-term production strategies and reservoir management, particularly in oilfields undergoing secondary recovery through water and gas injections, either simultaneously or alternately.

One example of application is evaluating the effects of different injection strategies or well unavailability. This framework enables the study of multiple scenarios much more efficiently, considering that forecasts can be reliable up to the medium term (scale of months).

2. METHODOLOGY

In this work, we exclusively utilize databases containing produced and injected volumes, represented as time series data with daily discretization and segmented by well, for both training and evaluating forecast methods. We employ supervised learning methods to establish the mapping between input and output variables, leveraging available data to extract knowledge.

When dealing with continuous output variables, the problem is classified as regression (Pandey et al. 2020). The primary human task in this process is to identify the algorithm capable of accurately representing the relationships between

input and output variables. Subsequently, the model is trained iteratively to minimize errors in output variables, a process known as the training step in the machine learning framework. Following training, a reserved portion of the dataset is utilized for validation, ensuring the model's performance is neither underfitted nor overfitted to the training set.

Our proposed solutions are based on multilinear and neural network regression models, which are at the forefront of a framework tailored to address time series data influenced by multiple concurrent time series, devoid of periodicity, seasonality, or stationarity. The algorithms are implemented in *Python*. Our workflow encompasses the following steps, each of which is detailed in subsequent sections:

(1) ETL (Extract, Transform, Load): External data sources are accessed and prepared for use within the program.

(2) Data Cleaning: Non-numerical values and outliers are filtered out, and data is sampled using mean values over a specified period.

(3) Dataset Reshaping: Time windows are applied to reshape the dataset, making it suitable for multivariate time series analysis.

(4) Dataset Splitting and Normalization: The dataset is divided into training, validation, and test sets based on temporal relationships. Subsequently, Gaussian normalization is performed, utilizing mean and standard deviation calculated solely from the training subset.

(5) Machine Learning Tuning: The architecture and hyperparameters of the regressors are defined and optimized.

(6) Forecasting: Trained regressors utilize production/injection history up to the reference date and future injection schedules as inputs to generate forecasts.

(7) Evaluation: During the training phase, evaluation metrics are computed using the validation subset to assess model performance and guide further tuning.

(8) Results Deployment: Once regressors are optimized, forecasts are generated using the test subset, and final results are deployed.

2.1 Data Conditioning

The initial steps in data conditioning involve exporting the production curves, loading them into the *Python* environment, formatting, and inspecting. Subsequently, the daily data can be aggregated into mean values for longer periods, such as weekly or monthly discretization. This flexibility is crucial to ensure database consistency with the reservoir dynamics and to facilitate working with the necessary time windows for effective forecasts.

The next crucial step is reshaping our dataset to suit the requirements of our forecast model, as illustrated in Fig. 1, which depicts an example of forecasting for an entire field. In this representation, blue boxes denote input variables for observations from previous time steps (denoted by i), while gray boxes represent estimated outputs for future time steps (denoted by k). Time t indicates the moment of the first estimate. The production variables Q_o, Q_g and Q_w represent oil, gas and water flow rates, respectively, while Q_{gi} and Q_{wi} denote injected gas and water rates.

A critical aspect of this task is selecting the “look back” and “look forward” windows, determining the number of past time steps used as inputs for the forecast and the number of future time steps estimated in each iteration. Evaluating different configurations to identify those with the best performance is a key focus of our research.

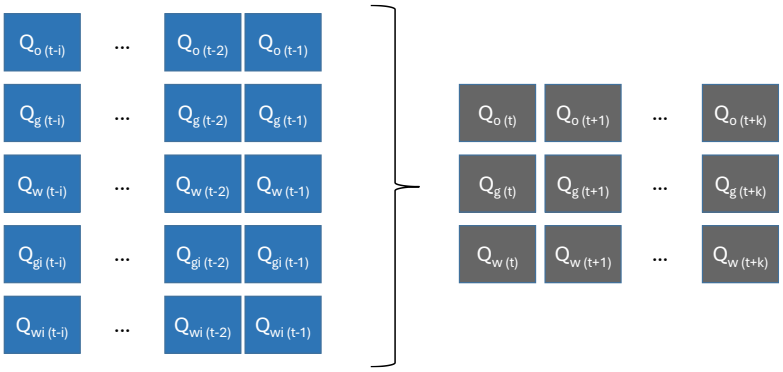


Figure 1. Reshaping the dataset for forecasting: blue boxes represent inputs, gray boxes the outputs for the regressors.

For a forecast segmented per well, each of the variables (Q_o, Q_g and Q_w for producers, and Q_{wi} or Q_{gi} for injectors) is considered to be in the range [t-1, t-i]. The outputs will represent the variables associated with each producing well in the interval [t, t+k]. Thus, the number of inputs and outputs for a problem with the number of producer wells n_{prod} and the number of injector wells n_{inj} will be given by:

$$n_{inputs} = i * (3 * n_{prod} + n_{inj}) \tag{2}$$

$$n_{outputs} = (k + 1) * (3 * n_{prod}). \tag{3}$$

We illustrate the dataset reshaping process in Fig. 2, following the previously described methodology. The figure is divided into two parts: the original dataset containing oil rates for three wells and the reshaped configuration intended for training and testing. The colors aid in visualizing the rearrangement for the first time step.

The obtained dataset is divided into training, validation, and testing subsets. It is crucial to note that, given the time series nature of the data, this division cannot be random. Data preceding our reference date are designated for training, while data following this date are allocated for validation and testing (Nielsen 2019).

Following dataset division, Gaussian normalization is performed. Mean and standard deviation are calculated solely from the training set, and these values are used to normalize the other subsets. This approach ensures that no influence from future information is carried into the learning process.

	Oil rates		
Timestep	P1 Qo	P2 Qo	P3 Qo
1	10	130	250
2	20	140	260
3	30	150	270
4	40	160	280
5	50	170	290
6	60	180	300
7	70	190	310
8	80	200	320
9	90	210	330
10	100	220	340

	Training Inputs (x_train)									Training Outputs (y_train)								
Timestep	P1 Qo t-3	P1 Qo t-2	P1 Qo t-1	P2 Qo t-3	P2 Qo t-2	P2 Qo t-1	P3 Qo t-3	P3 Qo t-2	P3 Qo t-1	P1 Qo t	P1 Qo t+1	P1 Qo t+2	P2 Qo t	P2 Qo t+1	P2 Qo t+2	P3 Qo t	P3 Qo t+1	P3 Qo t+2
Training only	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	10	20	30	130	140	150	250	260	270	40	50	60	160	170	180	280	290	300
5	20	30	40	140	150	160	260	270	280	50	60	70	170	180	190	290	300	310
6	30	40	50	150	160	170	270	280	290	60	70	80	180	190	200	300	310	320
7	40	50	60	160	170	180	280	290	300	70	80	90	190	200	210	310	320	330
8	50	60	70	170	180	190	290	300	310	80	90	100	200	210	220	320	330	340
9	60	70	80	180	190	200	300	310	320	90	100	110	210	220	230	330	340	350
10	70	80	90	190	200	210	310	320	330	100	110	120	220	230	240	340	350	360

Figure 2. Example of the dataset reshaping process.

2.2 Regression Methods

In this study, we employ linear regression and regularized variants. Linear regression predicts outputs as a linear combination of features, minimizing mean squared error (Pandey et al. 2020). We implement three variants: standard linear regression for initial exploration (Nielsen 2019), Ridge with L2 regularization to reduce overfitting, and Lasso with L1 regularization that zeros irrelevant coefficients, improving interpretability (Kalita et al. 2024). Parameter α controls penalty strength. All variants are evaluated comparatively.

We also employ artificial neural networks, specifically Multi-Layer Perceptrons (MLPs), which consist of interconnected layers of neurons that learn non-linear relationships through backpropagation—an algorithm that iteratively adjusts weights by computing loss function gradients (Braga et al. 2000). This enables MLPs to capture complex patterns beyond linear assumptions (Haykin 2008). We configure network architecture, including hidden layers, neurons per layer, and activation functions, through systematic experimentation.

2.3 Evaluation Metrics

We pay close attention to selecting appropriate metrics for evaluating the performance of predictors/estimators, especially when working with time series data. We choose the mean absolute percentage error (MAPE), the mean squared error (MSE), and the symmetric mean absolute percentage error (SMAPE). These metrics are calculated using the following equations:

$$MAPE = \frac{1}{n}\sum_{j=1}^{n}\left|\frac{y_j - \hat{y}_j}{y_j}\right| \tag{4}$$

$$MSE = \frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2 \tag{5}$$

$$SMAPE = \frac{1}{n}\sum_{j=1}^{n}\frac{|y_j - \hat{y}_j|}{\frac{(y_j + \hat{y}_j)}{2}} \quad (6)$$

where y_j represents predicted values and $\hat{y}_j$ actual values of the dependent variable (Rezende 2018, Kalita et al. 2024).

3. CASE STUDY

Our case study is centered around a simulated oilfield dataset generated from the UNISIM III compositional simulation model (Correia et al. 2020). This model represents a karst carbonate reservoir and was developed by researchers from the University of Campinas (UNICAMP) using public data from Brazilian pre-salt reservoirs supplemented with synthetic data.

The version of the UNISIM III model used in our study depicts an early stage of the field's development, as illustrated in Fig. 3. In this stage, wells are primarily concentrated in the top of the structure. Specifically, the configuration includes six oil-producing wells (P11, P12, P13, P14, P15, and P16), while a seventh well (P21) remains inactive in an isolated area. Additionally, there are seven injector wells, with three initially configured for water injection (I12, I14, and I15) and the remaining four for gas injection (I11, I13, I16, and I17).

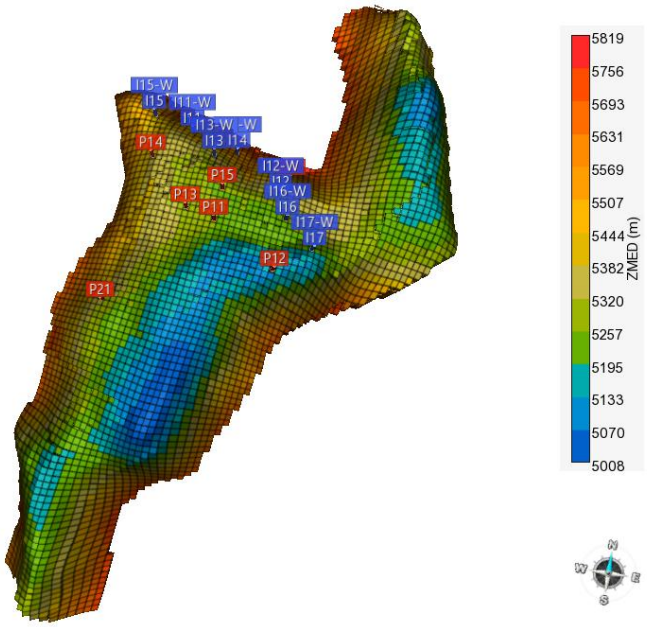


Figure 3. Representation of the UNISIM III reservoir simulation model. Based on data from Correia et al. (2020).

To construct a database for training and evaluating forecasting methods, we conducted a simulation of the UNISIM III model, extrapolated over a 20-year period. This simulation incorporated various changes in injection rate strategy throughout the timeframe to introduce richer data, reflecting real-life scenarios with operational inefficiencies and well/equipment unavailability.

Using this dataset, we explored and evaluated various configurations of estimators and different approaches to the forecasting problem, including:

- Forecast for the entire field or segmented by well: We assessed the performance of methods in both of these variants, recognizing their significance in practical applications;
- Data sampling (from originally daily rates): We aimed for a compromise solution, considering that sampling at shorter intervals provides greater granularity and more training samples but also increases cumulative error due to a higher number of forecast steps;
- Single-step or step-by-step (incremental) forecast: In the single-step approach, the predictor generates outputs for the entire defined look-forward window in a single prediction step (according to Eq. 3). Conversely, in the step-by-step approach, the forecast outputs are provided for each individual future time step through an iterative process. This iterative process involves calculating the next time step using inputs comprising *i*-1 past months of production/injection history (see Eq. 2), the planned injection for the first future step (from a previously given injection schedule) and the results of the production forecast from the previous step. This process continues until the desired production period is completed (at least six months, as defined). It is important to note that as we project further into the future, true historical data is

progressively replaced by previously forecasted data, which may amplify cumulative error. However, initial tests revealed vastly superior performance of the step-by-step approach, particularly due to its capability to incorporate injection schedules throughout the forecast period, leading us to adopt this approach exclusively;

- Look-back and look-forward windows: These parameters determine the number of simultaneous steps used as inputs and outputs for the predictors. The look-back window specifies the number of inputs for the estimator, influencing the time window captured, particularly relevant due to the dynamics of injections. Since we have adopted the step-by-step approach, the look-forward parameter remains constant at k=1 for subsequent evaluations;
- Time intervals for training and retraining: This evaluation focuses on determining the minimum historical data required for training to achieve satisfactory performance. Additionally, it examines the ideal frequency for retraining the estimator, considering the incorporation of new historical data and whether retraining should use incremental or fixed windows. This is crucial due to the non-stationary nature of time series data, where production rates typically decrease over time, and the susceptibility to concept drift, such as new water or gas breakthroughs;
- Machine learning methods, configurations, and hyperparameters: Various configurations and hyperparameters of linear regressions and neural networks are explored. For regressions (Lasso and Ridge), the primary parameter is the weight of the regularization term, α. The architecture of neural networks significantly influences performance and is chosen based on problem complexity and data availability. While the ideal network configuration may not be immediately apparent, starting with a simple network and gradually increasing complexity (Rubo et al. 2019) or using empirical equations based on input, output, and sample sizes – considering the undesired possibility of overfitting (Al-Bulushi et al. 2012) – can guide the selection process. In this study, we aim to balance these recommendations while determining the number of elements in a single hidden layer. We also experiment with different activation functions and use the *Adam* algorithm for training (Scikit Learn 2020).

4. RESULTS

We first evaluated production forecasts for the entire field, without individualization per well. This involved considering total production rates (Q_o, Q_g and Q_w) and injection rates (Q_{wi} and Q_{gi}). We conducted an evaluation employing a grid search approach for hyperparameter tuning. In our evaluation, we calculated metrics from a variety of predictors using an incremental training set and calculated the average values. Preferably, we considered metrics such as SMAPE and MAPE, as they are more suited to time series analysis, but we also computed MAE, and MSE.

We established limits for the observation windows, considering the time required to capture reservoir dynamics. This time was determined by multiplying the number of look-back samples by the sampling period. We found that the combination of a 10-day sampling period and 15-sample look-back window yields good outcomes, capturing events and offering greater stability in step-by-step forecasts.

We recommend working with incremental windows, where training utilizes all available historical data from the start of forecasting without discarding older data over time. Additionally, shorter retraining periods result in improved performance. Therefore, we suggest retraining the model on an annual basis.

When assessing the estimators, we explored numerous configurations. Table 1 compiles evaluation metrics for the most relevant estimator configurations, including simple linear regression, Ridge regression with various alpha values, and neural networks with different numbers of elements in the hidden layer. Notably, among the regression models, Ridge regression consistently outperformed Lasso regression.

The simplest method, linear regression, outperformed the other methods in this case. This superiority may be attributed to the relatively smooth variations in total field curves, which can be adequately captured by linear relationships, as suggested by Kubota and Reinert (2019).

Table 1. Compilation of evaluation metrics for selected estimators in the simulated full field case study.

Metric	LR	RR			NN		
	-	α=0.05	α=0.20	α=0.40	10 el.	20 el.	70 el.
SMAPE	0.105	0.116	0.125	0.128	0.149	0.146	0.148
MAPE	0.298	0.328	0.301	0.325	0.472	0.367	0.413
MAE	0.027	0.033	0.041	0.046	0.086	0.074	0.072
MSE	0.002	0.002	0.003	0.004	0.015	0.010	0.010

Based on these results, we have chosen for further evaluation the simple linear regression model and the neural network with 20 elements in the hidden layer. The activation function employed was *identity*, selected in our tuning process. We then conducted forecasts over an observation period and compared them with the reference values obtained from the simulation model. To illustrate that for a period, we set t=0 to represent a period of constant gas injection

followed by a subsequent increase in water injection rates a couple of days later, as shown in Fig. 4, representing an injection schedule that we can define for the forecast and feeding the model with the injection rates at each time step. Forecasted results for the next six months (18 steps, with a 10-day sampling interval) are presented sequentially for a Neural Network with 20 neurons (Fig. 5) and linear regression (Fig. 6).

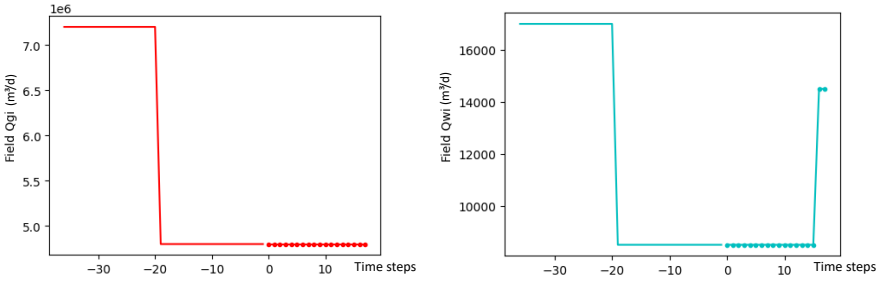


Figure 4. Injection schedule in our example for the full field simulated case. Each time step corresponds to 10 days.

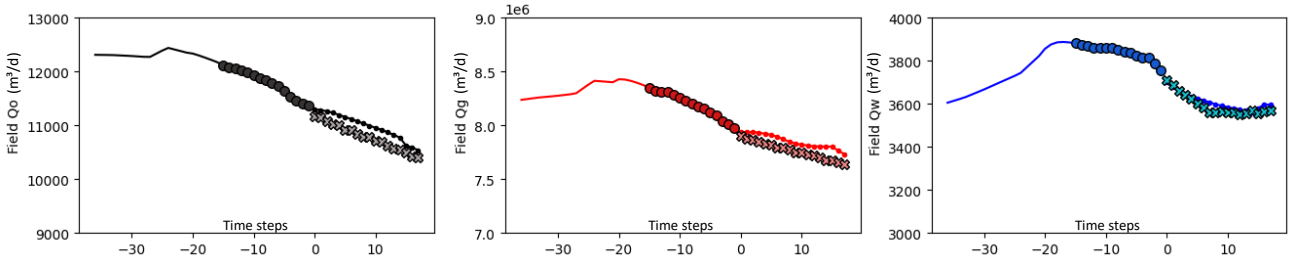


Figure 5. Production rates forecasted by NN; example for the full field simulated case. Forecast performed for the next 180 days (x), using as inputs the previous 150 days (○). Each time step corresponds to 10 days.

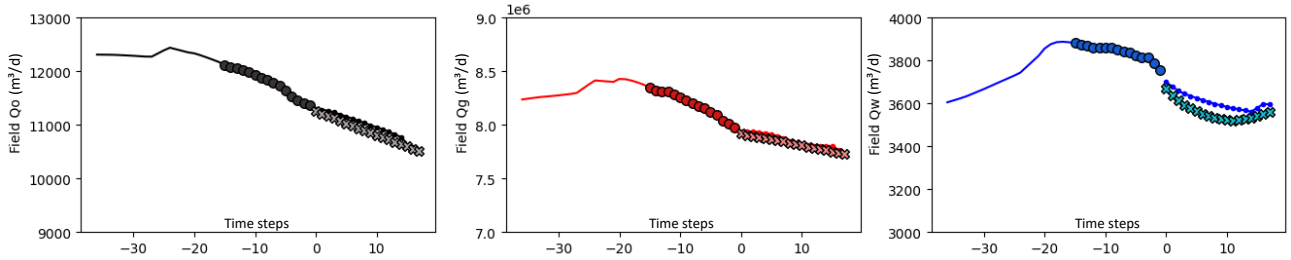


Figure 6. Production rates forecasted by linear regression; example for the full field simulated case. Forecast performed for the next 180 days (x), using as inputs the previous 150 days (○). Each time step corresponds to 10 days.

In the next stage, we evaluated the methodology in a case study segmented per well, starting from an estimator configured based on the results of the previous case study. Again, we observed that a longer training history tends to yield better results. Similarly to the previous case, we also explored the optimal choice for the sampling and look-back combination, obtaining this time the values of 10 days and 25 samples, respectively.

Upon evaluating the estimators, linear regression did not yield satisfactory results, possibly due to the increased complexity in the relationships among wells, which exhibit more prominent non-linearities. Conversely, Ridge regression emerged as the best-performing method across all metrics, as evidenced in Table 2.

Table 2. Compilation of evaluation metrics for selected estimators in the case study of an oilfield segmented per well.

Metric	**LR**	**RR**			**NN**		
	α=0	α=0.05	α=0.20	α=0.60	10 el.	20 el.	40 el.
SMAPE	0.322	0.158	0.151	0.147	0.235	0.215	0.214
MAPE	1.446	0.186	0.272	0.182	0.558	0.444	0.364
MAE	0.806	0.196	0.179	0.176	0.286	0.265	0.264
MSE	6.900	0.108	0.081	0.072	0.163	0.144	0.157

Finally, we analyze the curves by forecasting a period of 12 months into the future. The results obtained using our best estimator (Ridge regression with α=0.6) are presented in Fig. 7, illustrating the performance across the six producing wells in the model. The estimator demonstrates a satisfactory capability to reproduce the major trends observed in the data, including not only oil production but also the increases and decreases in gas and water production that follow the changes in controlled injections. Some divergences between predicted and observed results arise from cumulative errors inherent in sequential step-by-step forecasting. Additionally, the relatively simple data-driven models face inherent limitations in accurately representing the complex reservoir dynamics. The underlying reservoir model comprises thousands of grid cells, each characterized by multiple rock and fluid properties that evolve over time, particularly through depletion and dynamic changes in fluid saturations. These complexities make it challenging to achieve a purely data-driven representation without incorporating additional model refinements.

We emphasize that injection rates for forecast periods are predefined and supplied to the model at each time step. This framework enables simulation of different injection schedules to evaluate alternative reservoir management strategies. Furthermore, these schedules can be optimized to maximize oil recovery, what is a key application of our work. By providing rapid evaluations, the methodology serves as an efficient screening tool: promising strategies identified through data-driven forecasting can be further refined and validated using detailed numerical simulation.

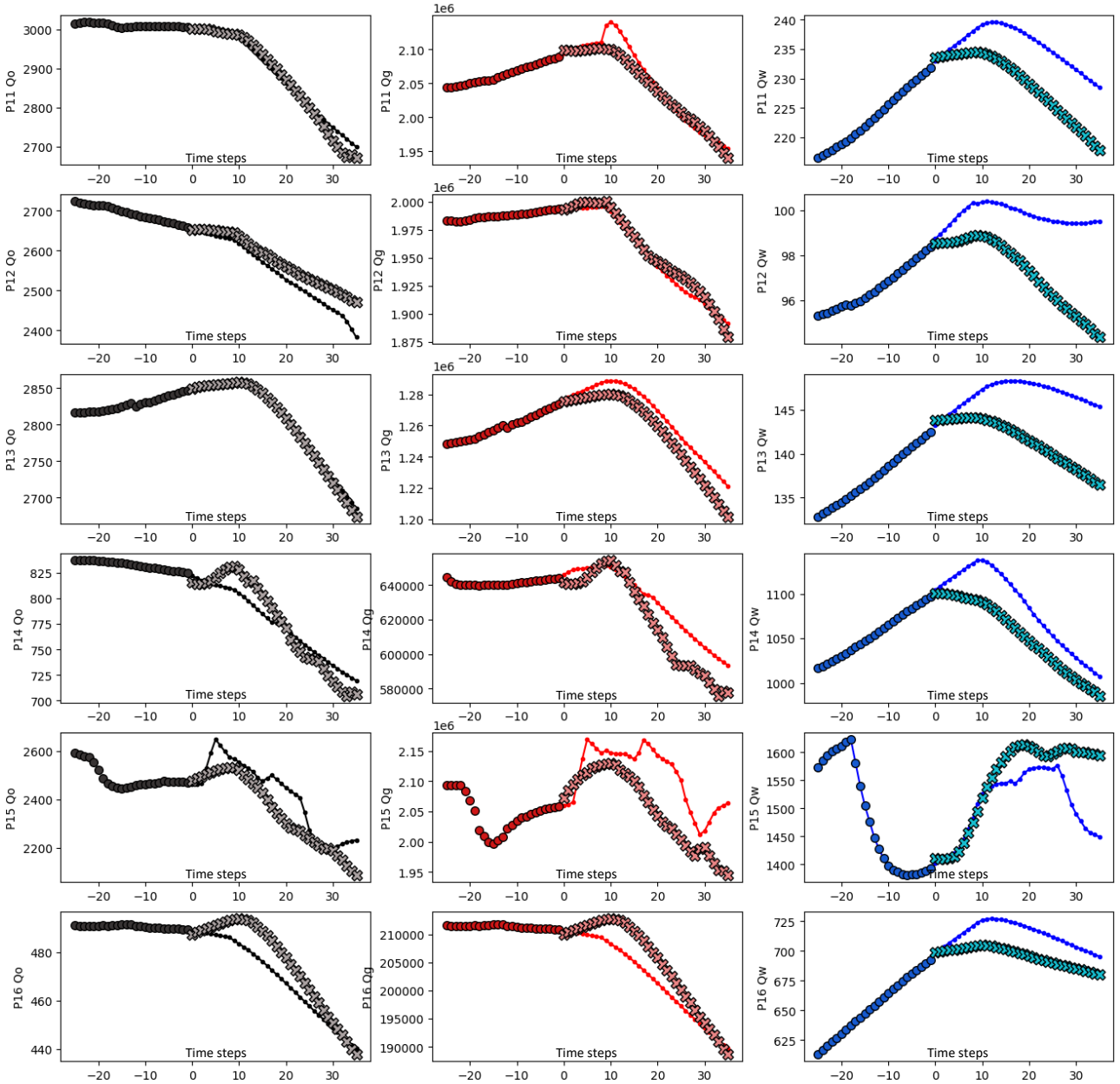


Figure 7. Production rates forecasted by Ridge regression for the example of forecast segmented by well. Forecast performed for the next 360 days (x), using as inputs the previous 250 days (○). Values in m³/d. Each time step corresponds to 10 days.

5. CONCLUSIONS

This work presents the development of a data-driven workflow for production forecasts in oilfields. Our approach encompasses crucial steps of data conditioning and the evaluation of machine learning methods, specifically linear regression and neural networks. The results obtained are promising for establishing a reliable predictor capable of delivering rapid responses, solely based on past production data and scheduled injection rates, without requiring information related to geology, rock/fluid interactions, or individual well and production facility details.

A novel aspect we introduced is the ability to address temporal and volumetric relationships between individual wells. Additionally, the framework enables forecasting of not only oil but also gas and water rates, along with the provision of injection rate schedules for upcoming steps. These features enhance the versatility of the framework, rendering it valuable for reservoir management and evaluating multiple production strategies at minimal computational costs and running times.

We emphasize our aim to provide reliable forecasts for a minimum period of six months, deemed suitable for practical applications such as operational planning and short- to medium-term reservoir management decisions. The results of the case studies are promising, with the models successfully capturing general production tendencies even in challenging scenarios. However, we acknowledge room for improvement when dealing with longer forecasts and transient behaviors.

Furthermore, we highlight the significant sensitivity to configurations and hyperparameters of the regression methods, underscoring the importance of obtaining representative and reliable databases as the fundamental basis for any data-driven approach.

Future work will extend this study to real-world applications and address current limitations. Given that the present work employs simulated data as a proof of concept, which lacks the noise and imperfections inherent in field observations, subsequent studies will incorporate actual reservoir data and develop methods to handle noisy and incomplete measurements. Additionally, hybrid approaches combining physics-informed constraints with data-driven methods will be explored to enhance model accuracy and robustness when dealing with real-world complexities.

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